

CITS2402 Introduction to Data Science

Semester 2, 2026

Project

Marks: 60 (team work: 50, Individual report: 10)

Weight: 20%

Team members: 2-3

Team Registration: 11:59pm, Thursday 10th September 2026 [Submission Link: <https://forms.cloud.microsoft/r/VtTA3VXw5d?origin=lprLink>]

Project Submission deadline: 11:59pm, Friday 2nd October 2026 [Submission on LMS]

1 Aim

This project aims to explore the similarities and differences in a demographic feature of your choice by comparing data from the most recent Australian Census (2021) with data from the 2023 New Zealand Census.

From the data, you may choose what census topic you are most interested in, provided it is not the one that is addressed in the lecture or lab case studies (such as age, number of children, or travel to work, etc.). For best results, you should also choose a topic with multiple categories (not, for example, binary categories). You may focus on particular categories of interest. You should use appropriate ways of visualisation of the data that best demonstrates the similarities and differences in your conclusions.

To focus on the project, it may be helpful to frame it as a question you seek to answer. Ensure your question is complex to demonstrate your data science skills in depth. You should clearly state in your opening paragraphs what it is that you are seeking to answer.

When selecting your topic, choose data that lends well to a more comprehensive analysis and graphical comparison. Simple comparisons, such as population size, are unlikely to achieve high marks. Review several alternatives before settling on your topic to ensure it offers rich data for analysis and meaningful insights. Choose a topic that interests you and allows for a comprehensive and engaging exploration using the available census data. It is recommended to explore few topics initially to see which ones lead to an interesting investigation and determine whether the available data is sufficient to address your question.

Here are some context **examples** to inspire your choice:

- Comparing urbanisation trends in both countries for their census years.
- Examining the impact of migration on cultural diversity in both countries.
- Analysing the relationship between education levels and life satisfaction in both countries.
- Comparing housing or health indicators and their impact on overall well-being in both countries.

It is important to understand from the beginning that, while programming is important, this assignment is not merely a “coding exercise”. Equally significant are the context and rationale behind your work, the sound and replicable use of data, and the clear and compelling presentation of your results.

2 Learning outcomes

This assignment demonstrates competencies in:

- sourcing information from (authoritative) public data repositories.
- extracting and cleaning information needed to answer a question about the data.
- analysing and interpreting the data.
- visualising data to aid understanding and communicate results.
- writing comprehensive and informative scientific reports to communicate your findings.
- clearly stating your assumptions about the data and its context, along with the rationale and assumptions behind any data preprocessing steps.

3 Authorship

The project must be done in groups of two or three students.

Each student’s name and student number must be provided in the declaration at the top of the project template

`CITS2402-Project-template.ipynb`.

Only one copy should be provided to mark. In case multiple students make the submission then anyone will be graded and penalty will be applied.

The suffix “-template” should be replaced with the corresponding student numbers. For instance, if you are doing your project in a group of two members then you should rename your file as `CITS2402-Project-STDNO1-STDNO2.ipynb`, where ‘STDNO1’ and ‘STDNO2’ are the corresponding student numbers involved in the submission.

The submission must be the students’ own work. Any material used in the assignment from other sources must be clearly stated and referenced.

It is recommended that you download a backup of your final completed submission directory for your own records.

4 Data

Finding the appropriate data is part of the exercise. It is possible that the Census of both countries may not provide the data in the same way. It is recommended that you begin

searching for your data early on, using the Australian Bureau of Statistics (ABS) and Stats New Zealand. The metadata in the spreadsheets should be used to identify the relevant tables for your investigation. Only the individual tables in CSV format (not all the data) should be used in your submission.

Your report should clearly explain how you located the relevant data. This explanation should be detailed enough to allow the reader to replicate your steps and obtain their own raw data to test your code.

Your code should only need to access the file system for reading. You should not save images, or write files, etc.

Data Collection: To gather data, start by exploring the “DataPacks” under the “General Community Profile” on the Australia Bureau of Statistics website:

1. Go to the Census page.
2. Navigate to Find Census data.
3. Proceed to Census data tools.
4. Finally, go to DataPacks.

From there, for 2021, select “Australia” under “geography” and select “General Community Profile” under “DataPack type”. You will find a ZIP file containing metadata and data. Afterwards visit Stats New Zealand website and “Aotearoa Data Explorer” to download relevant data.

5 Submission

Your team submission should be made by one project member and it consists of:

1. The Python notebook `CITS2402-Project-STDN01-STDN02.ipynb`, including all explanations and code. It should contain headings and explanations in markdown cells and executable code in Python cells.
2. The PDF version of the Python notebook. Both files should be named following the instructions above.
3. Any data files that you use to run the code. Data files should not be more than 5MB.

Your individual submission consists of:

4. Each student is required to submit a confidential pdf file `Project-Reflection-STDN` of working in a team, work distribution in the project and how they rank other project members. You are required not to share this report with anyone.

Submit your files to LMS **as a ZIP file** before the due date and time. You can submit them multiple times. Only the latest version will be marked. Pay attention if you are submitting all requested files when making a new submission. Your submission will follow the rules provided

in LMS.

Before submitting your project, you must ensure it runs without errors in the notebook (Google Colab environment). This is to avoid markers having local problems with libraries you may choose to use. Therefore, your code should run seamlessly without having to install any package on Google Colab. The Colab environment has embedded essential data science libraries (numpy, pandas, matplotlib, etc). Your mark will be zero if marker cannot run your code in Google Colab.

Important:

- It is your responsibility to ensure that you follow submission guidelines. In case you submit files in incorrect format or missing data, you will receive a zero mark.
- Failing to submit any of the required files will result in a zero mark. You should include all data files you are using.
- The data files you are using should be clearly specified. The marker should be able to download the same data files from the website(s) and run your analysis.
- You should provide comments on your code.
- By submitting your assignment, you acknowledge you have read all instructions provided in this document and LMS.
- There is a section in your LMS -> Project - Updates and Clarifications, where you will find updates or clarifications about the tasks when necessary. It is your responsibility to check the page regularly. Also you are suggested to keep an eye on discussion forum on LMS to updated QnA about the project.
- You will be assessed on your critical thinking and process, not only on your results. You should demonstrate you understand the concepts involved.
- Your answer must be concise. You will be graded on thoughtfulness. If you are writing long texts, rethink what you are doing. Probably, it is the wrong path.
- You can ask in the lab or during consultation if you need clarification.
- You should be aware that some algorithms can take a while to run. A good approach to improve Python's code speed is using the vectorised forms as discussed in lectures. In this case, it is strongly recommended that you start your project soon to accommodate the computational time.

6 Code

The code will be executed with a fresh kernel for marking, so (as usual) you should ensure it runs with a clean kernel before submission.

Any supporting data must be in the same directory. Data files should not be more than 5MB.

It is recommended that development is done in the notebooks - in the past students' code fail to run due to pasting from other environments.

7 Rubric

The project involve analysing data. You should present a well-organised report and aim to write concisely. The project will be marked for clarity and professionalism of both the exposition and the coding.

Please read the project instructions carefully when preparing your code and report.

It is recommended that you structure the report in a way that is consistent with the data science lifecycle. You should use headings to help structure your report.

Consider these aspects as examples of what is expected:

- Plots and figures: Your report should include appropriate and well-presented visualisations that are meaningful to the analysis. All your diagrams/plots should have proper titles, axis labels, values, legend, etc., to help the reader understand what you are plotting.
- Your report should include the correct use of the data science lifecycle steps, the interpretation of the results obtained from these steps and the descriptions of assumptions used in the analysis.
- Presentation of results: Describe the results of your analysis and their interpretations. **Software output is not a valid output.** You must format and present your answer appropriately (tables, graphs, etc.). You should not add irrelevant information when presenting the results.
- Discussion: Based on your results, describe the conclusions of your analysis.
- Overall: Do not report out a long list of numbers if you do not explain what they are and what they mean. Do not generate plots after plots without explaining what they were doing or their purposes. Instead, think about the best way to display/illustrate your approach. Pay attention to all details, including the graph labels, presentation quality, and clarity.

Markers will pay attention to the following components (roughly equal weighting).

Context and data:

- Adequate context has been provided to understand the question and why it is important.
- The question you seek to understand is clearly stated.
- It is clear what data is used and its provenance. Instructions allow the reader to easily source the data (to make the work replicable).
- Complete context and information about the data (e.g. the unit of measure, description of the categories in the topic, etc.) are provided.
- Relevant differences between the data from different years and assumptions you have to

make for comparison are clearly described.

- If you are extracting only part of the data, your code should be accompanied by a brief description of what you are extracting and why.

Data lifecycle, structure, and presentation:

Your data cleaning steps should be accompanied by a brief description of any steps you took to transform the data from its raw form into usable form.

- The route from the data to the results is clearly set out, and steps are explained.
- It is clear what format the raw data took, what is extracted and why.
- Any data cleaning and conversion is clearly and concisely outlined (with rationale, assumptions or justifications).
- The processing or analysis necessary to extract and compile the results is clearly explained.

Results, visualisation and conclusion:

- The results are clearly stated and connected back to the original data and assumptions.
- Appropriate and informative choices are made for visualisation(s) (plots).
- The visualisations are clearly and professionally presented.
- Conclusions are connected to relevant features of the visualisations.

Coding:

There is no single “right” way to write the code. However, the following should be considered:

- Code is clear and easy to read and comprehend. Considerations should include the use of meaningful variable names, the use of comments and/or docstrings for key steps/blocks (you do not need to comment every line; this tends to obscure the key steps), and the use of functions.
- Code is appropriately *concise*. Code should not be pared down to a bare minimum at the expense of clarity and readability. However, you should try to avoid unnecessary extraneous code.
- Code is reasonably efficient. (It is not necessary to achieve ultimate efficiency at the expense of writing clear, logical code. However, you should avoid obvious unnecessary inefficiencies.)
- Code is well *structured*. Functional decomposition is used to separate tasks into meaningful components.
- Ensure you avoid repeating unnecessary blocks of code by using functions. Additionally, refrain from hard-coding values directly within functions; instead, use function arguments to pass these values.

Professionalism and Challenge:

- Overall, the report forms a compelling and illuminating narrative.
- The report is not unnecessarily long or repetitive and provides all the information completely but concisely.
- The report reveals aspects of the data that are not trivially obvious.
- The report is of a quality that an employer would be comfortable showing to a client.

Remember your report should detail how you obtained the data and the coding steps used to create the subset for your project. That is, no manual data handling. The marker should be able to download the same data according to your details and obtain the same subsets and results you obtained when running your code.

Individual Reflection Report:

Each project member needs to submit reflection report and should reflect on critical reflection, self-awareness, and learning from the teamwork experience. It should also include what other members contributed and how you rate your team members' performances. Your report will be checked for:

- Description of roles and contributions
- Reflection on team dynamics
- Challenges and problem solving
- Learning and personal growth
- How truly you reflect on each team member's performance including yourself.

8 Plagiarism and penalty on late submissions

See the EVA Policy about late submission of project <https://www.uwa.edu.au/students/your-studies/exceptional-variation-of-assessment>

Plagiarism: In accordance with University Policy, you certify that all work submitted for this assignment is your own and that all material drawn from other sources has been fully acknowledged.